LCSC Cable - reduced RFQ

RFQ-CABLE-02-ECONOMY | 09 November2026

QUOTATION ONLY - NO ORDER OR PRODUCTION RELEASE

Four standard cable types for10 laboratory instrument sets. Base60cables; alternate70cables with optionalTMP117. This current reduced request supersedes the13-line all-custom comparisonRFQ-CABLE-01.

Type	Base quantity	Alternate TOTAL with TMP
H01	10	10
H02	10	10
H03	10	10
U04	30	40

U04 is one shared reference,30 OR40pieces, not30+40. Encoder and probe-jack soldering will be done locally. No custom encoder, jack, SH gauge transition, external probe, power, cutoff or USB subassemblies are included in this reduced supplier quote.

No special common outer jacket or overmould requested for internal cables. Each wire remains insulated. Ordinary flexible stranded copper and simple cavity numbers are required. Quote any lower-cost compatible socket with exact manufacturer/MPN/drawing separately; approval is still required.

Please itemize unit/setup/first-article/shipping/tax costs and lead time. No first article or production is authorized. Internal purchasing and estimated budgets are intentionally not attached to this supplier packet.

Common requirements and connector assignment

Flexible wire-path length between rear wire exits of connector housings, or solder joints/terminal shoulders for soldered ends. Includes fan-outs, branches and transitions along that path, excludes connector bodies and contacts. Supplier must state exact datum and physical tip-to-tip length on approval drawing; do not silently interpret as cut-wire length. 300mm is a quotation/service-loop proposal, actual routed fit unqualified.

100percent continuity and unintended-short check, standard cavity labels and crimp pull/retention controls per connector specification. Supplier provides mating/wire-entry views and cavity1. No recurring special-label/custom-jacket/overmould operations. No customer crimping.

XH2.50mm and independent header sockets2.54mm are distinct. JST XHP-2/3/4/6 + SXH-001T-P0.6 are the requested PCB-side family. U04 sockets use a documented Harwin M20-1060100 + M20-1160042 reference for0.64mm square male posts. Supplier alternatives need approval.

Provide mating-face and wire-entry drawings, with cavity1 and key/latch. Pin maps are logical cavity numbers; do not mirror based on looking at the other end.

U04 pin	LCD / BH1750 / optional TMP	Encoder after local soldering
1	GND	ENC_A
2	+3V3_A	ENC_B
3	I2C_SDA	ENC_SW
4	I2C_SCL	GND -> encoderC andSW2

U04 carries only numbers1..4. Do NOT mark a fixedGND/VCC function on every U04: encoderGND is pin4, while I2C peripheralGND is pin1. Each flying socket must remain individually movable.

Local catalogue probe pigtail, not supplied in this RFQ: B/J4.1 +3V3_B ->TIP; B/J4.2 NTC_RAW ->SLEEVE; ring isolated. SLEEVE IS NOT GND. No independent-cutoff requirement is waived by excluding it from this cable order.

H01 - A-B POWER

Base quantity 10 | alternate total 10 | RFQ ONLY

End A: A/J1 | JST XHP-2 | JST SXH-001T-P0.6

End B: B/J1 | JST XHP-2 | JST SXH-001T-P0.6

Wire 22AWG | L300+/-10mm between rear wire exits; supplier drawing must state exact datum

A cavity	B cavity/socket	Purpose	Colour / marking
1	1	GND	black
2	2	+5V_AB	red

Standard construction

Standard individually insulated flexible stranded copper wires, ordinary colours and cavity labels at each end. No custom common outer jacket, overmould, special tube or workshop soldering. Optional light bundling fitted during integration; retain full insulation, wire gauge and crimp quality.

Approval points

Electrical map fixed; supplier drawing, OD/bend radius and real routed service slack to approve

No custom soldering or special jacket is requested. Preserve contact/wire compatibility and normal crimp/continuity quality.

References: https://www.jst-mfg.com/product/pdf/eng/eXH.pdf | https://www.jst-mfg.com/product/pdf/eng/eXH.pdf

H02 - A-B I2C

Base quantity 10 | alternate total 10 | RFQ ONLY

End A: A/J6 | JST XHP-3 | JST SXH-001T-P0.6

End B: B/J6 | JST XHP-3 | JST SXH-001T-P0.6

Wire 26AWG | L300+/-10mm between rear wire exits; supplier drawing must state exact datum

A cavity	B cavity/socket	Purpose	Colour / marking
1	1	GND	black
2	2	I2C_SDA	blue
3	3	I2C_SCL	yellow

Standard construction

Standard individually insulated flexible stranded copper wires, ordinary colours and cavity labels at each end. No custom common outer jacket, overmould, special tube or workshop soldering. Optional light bundling fitted during integration; retain full insulation, wire gauge and crimp quality.

Approval points

Electrical map fixed; supplier drawing and actual service fit pending

No custom soldering or special jacket is requested. Preserve contact/wire compatibility and normal crimp/continuity quality.

References: https://www.jst-mfg.com/product/pdf/eng/eXH.pdf | https://www.jst-mfg.com/product/pdf/eng/eXH.pdf

H03 - A-B CTRL

Base quantity 10 | alternate total 10 | RFQ ONLY

End A: A/J7 | JST XHP-6 | JST SXH-001T-P0.6

End B: B/J7 | JST XHP-6 | JST SXH-001T-P0.6

Wire 26AWG | L300+/-10mm between rear wire exits; supplier drawing must state exact datum

A cavity	B cavity/socket	Purpose	Colour / marking
1	1	GND	black
2	2	PWM0	white
3	3	ENABLE	red
4	4	FAULT_N	yellow
5	5	PWM1 / B pad NC	blue
6	6	DIN0 / B pad NC	green

Standard construction

Standard individually insulated flexible stranded copper wires, ordinary colours and cavity labels at each end. No custom common outer jacket, overmould, special tube or workshop soldering. Optional light bundling fitted during integration; retain full insulation, wire gauge and crimp quality.

Approval points

Positions5/6 must still be wired through, even though PCB B temperature leaves those pads NC; no commoning to GND

No custom soldering or special jacket is requested. Preserve contact/wire compatibility and normal crimp/continuity quality.

References: https://www.jst-mfg.com/product/pdf/eng/eXH.pdf | https://www.jst-mfg.com/product/pdf/eng/eXH.pdf

U04 - Universal XH4 to four individual female sockets

Base quantity 30 | alternate total 40 | RFQ ONLY

End A: A/J2 LCD, A/J5 BH1750, A/J3 encoder; optional A/J4 TMP117 | JST XHP-4 | JST SXH-001T-P0.6

End B: Four independently movable female0.64mm-square-post sockets, numbered1..4 | 4 individual Harwin M20-1060100 female1-position housings | 4 Harwin M20-1160042 female crimp contacts

Wire 26AWG | L300+/-10mm between rear wire exits; supplier drawing must state exact datum

A cavity	B cavity/socket	Purpose	Colour / marking
1	1	GENERIC CAVITY 1	supplier standard; number authoritative
2	2	GENERIC CAVITY 2	supplier standard; number authoritative
3	3	GENERIC CAVITY 3	supplier standard; number authoritative
4	4	GENERIC CAVITY 4	supplier standard; number authoritative

Standard construction

Standard individually insulated stranded copper wires. Number1..4 permanently at BOTH ends; no fixed GND/VCC function labels because encoder has a different map. Four independent1-position female sockets, ordinary crimp construction. No custom jacket, gauge transition, soldering or overmould. Harwin parts are dimensional/reference proposal; quote exact documented compatible lower-cost MPN separately if cheaper.

Approval points

30pcs base:10 LCD +10 BH1750 +10 encoder. Quote40pcs as an alternative total if TMP117 is populated; NOT30+40. Encoder sockets will be cut off locally, stripped and soldered with a common-ground bridge by the integrator (user accepted). TMP117 uses retail QW-4C male tails with numbered universal female sockets. No direct custom XH-SH needed.

No custom soldering or special jacket is requested. Preserve contact/wire compatibility and normal crimp/continuity quality.

References: https://www.jst-mfg.com/product/pdf/eng/eXH.pdf | https://www.harwin.com/products/M20-1060100